BIOCHEMISTRY CUMULATIVE EXAM

October 5, 1991

Chromosome Structure

1. (20) If a single DNA molecule has a molecular weight of 2×10^{11}, about what length would it be in a fully extended double-helical conformation? (A sugar-phosphate-base monomer has a molecular weight of about 330, and there are 34 Å per turn of the double helix.)

2. (10) Compare your answer to #1 to an estimated length of an average cell. What "compaction ratio" (foreshortening) is needed to fit the DNA into the cell?

3. (20) Histones help to provide one level of DNA packaging. Tell what you know about the properties of histones as proteins. About how many kinds are there? Do you expect them to have high, low, or neutral isoelectric points, and why?

4. (30) Describe what you know about the structure of DNA-histone complexes.

5. (10) Name three scientists who have worked on histone or histone-DNA complexes.

6. (10) Approximately what "compaction ratio" (foreshortening) do histones provide for DNA? How does this compare with the ratio needed to pack a chromosome? Draw appropriate conclusions.